Investigating the Role of HbA1c Level in Predicting Diabetes Using Different Machine Learning Models

CBIO313: Machine Learning & Data Mining

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Presented by: Abdalmajed Kadry | ID: 221001002 | Spring 2026

INTRODUCTION

Diabetes is a chronic condition with rising global impact.

HbA1c is a key biomarker for long-term blood glucose control.

This project uses a dataset of 100,000 individuals to explore how well HbA1c predicts diabetes alongside other clinical features.

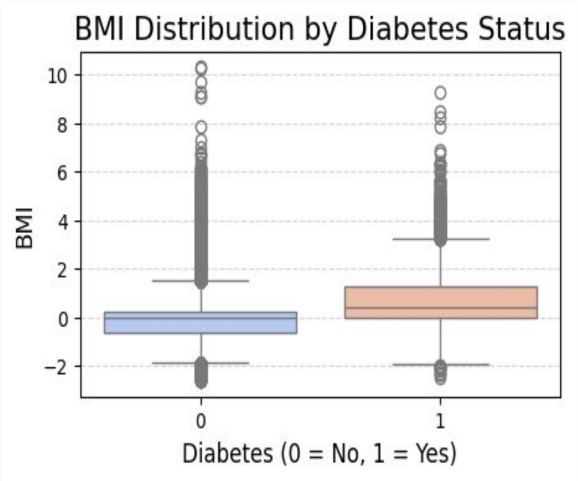
METHODOLOGY & RESULTS

1. Data Preprocessing

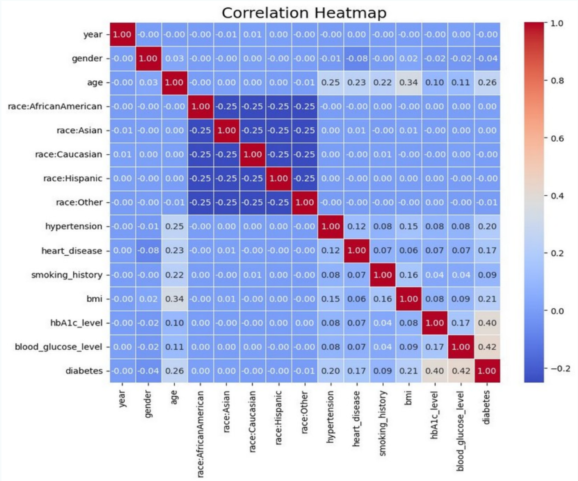
- Checking null values
- Checking duplication
- Encoding Gender & Smoking History
- Standardization (StandardScaler)
- 80 / 20 Train–Test Split (random_state=42)

METHODOLOGY & RESULTS

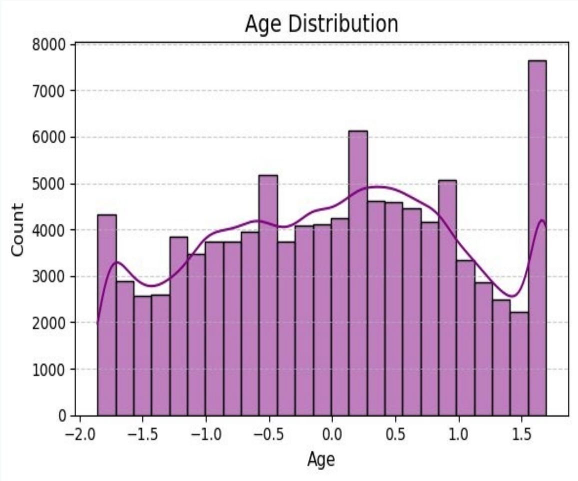
2. Exploratory Data Analysis



BMI Distribution by Diabetes Status



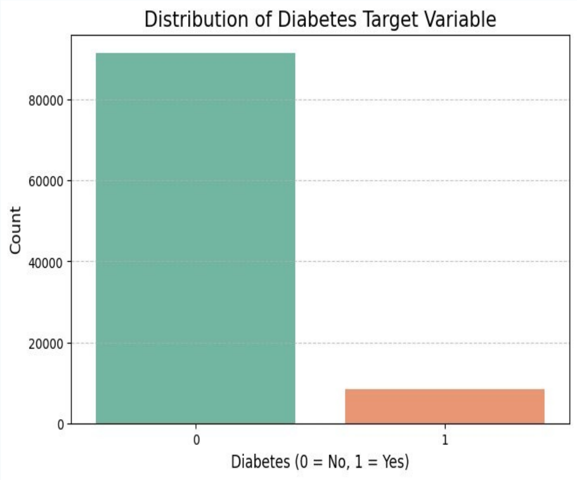
Correlation Heatmap



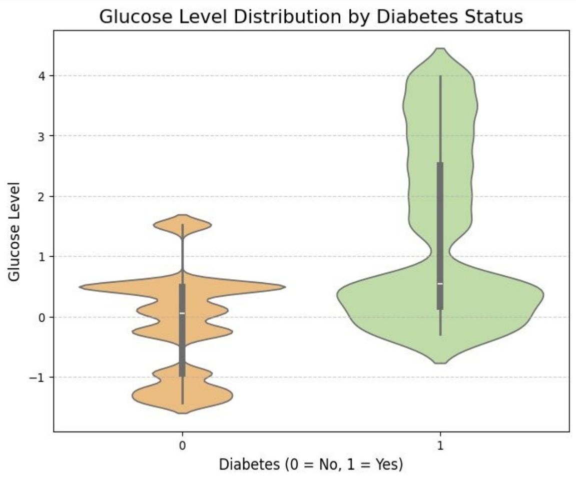
Age Distribution with KDE

METHODOLOGY & RESULTS

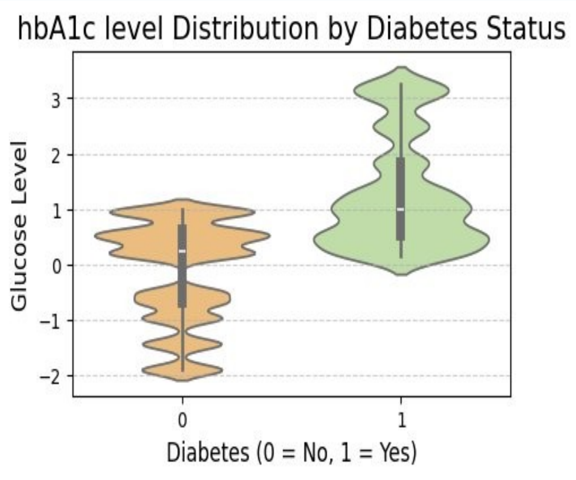
2. Exploratory Data Analysis



Target Variable Distribution



Glucose Level by Diabetes Status



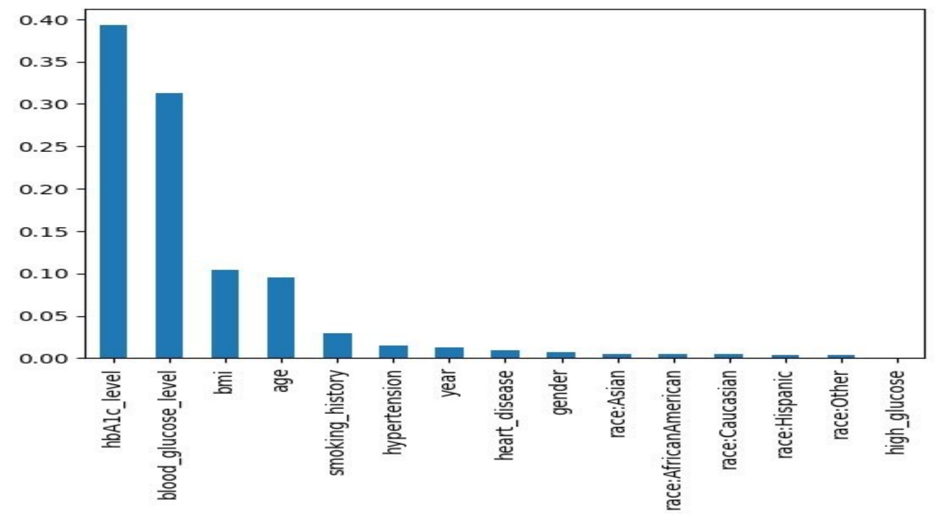
HbA1c Level by Diabetes Status

METHODOLOGY & RESULTS

3. Feature Engineering & Feature Selection

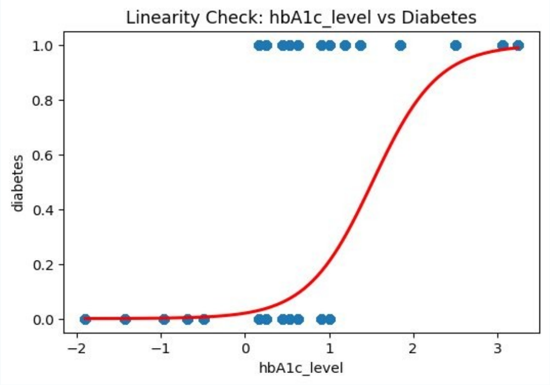
New binary feature created:
high_glucose = 1 if blood_glucose_level > 140 (high risk), else 0

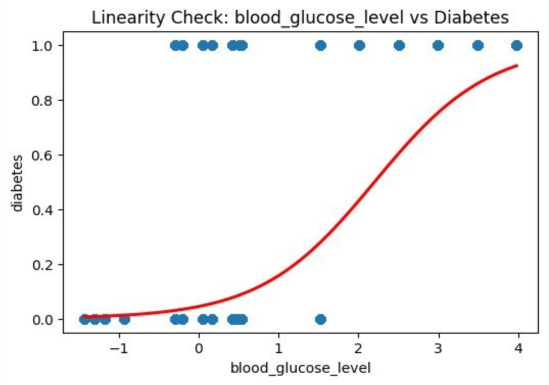
Feature selection via Embedded Method:
Random Forest importances → Top 5 features selected for final models.

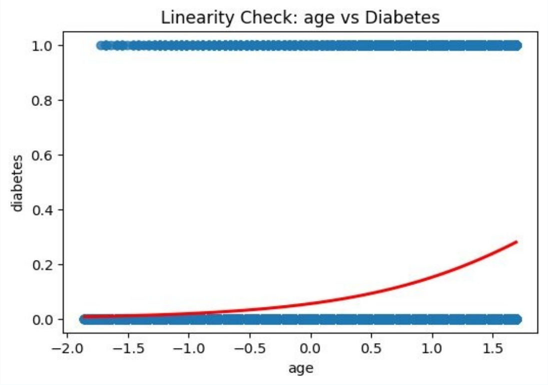


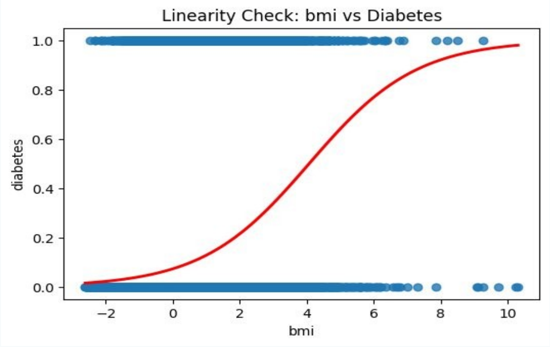
METHODOLOGY & RESULTS

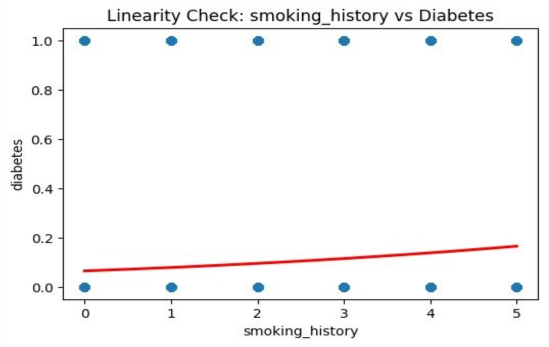
3. Feature Selection — Linearity Checks







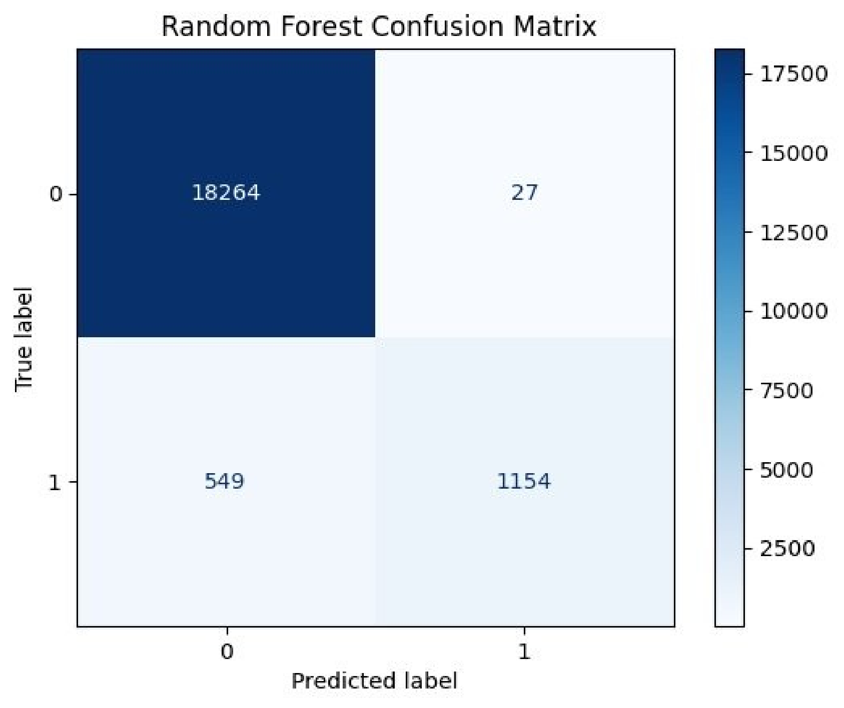


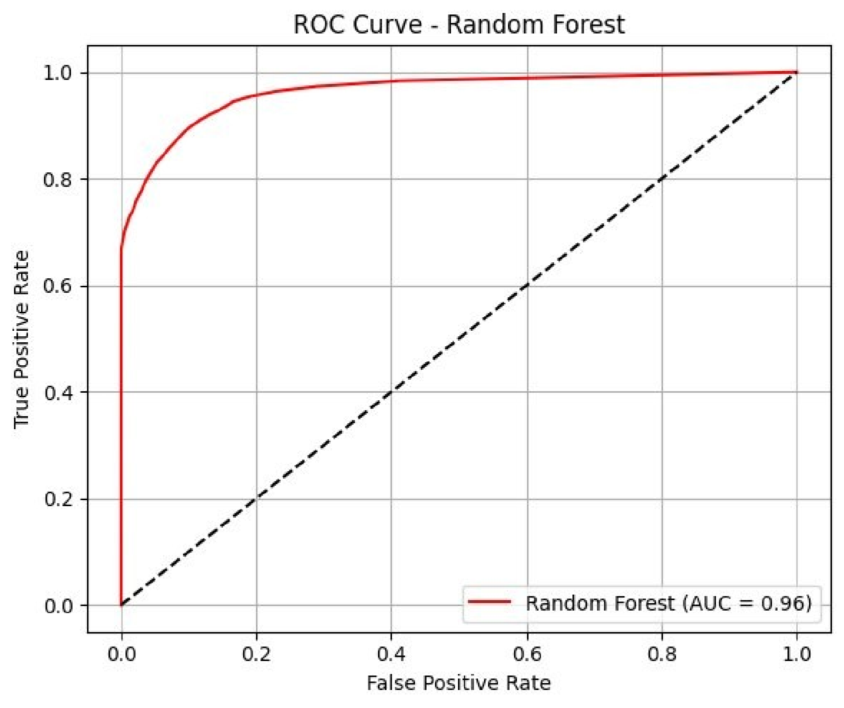


Logistic regression trend plots verify monotonic relationship between each feature and the diabetes target.

METHODOLOGY & RESULTS

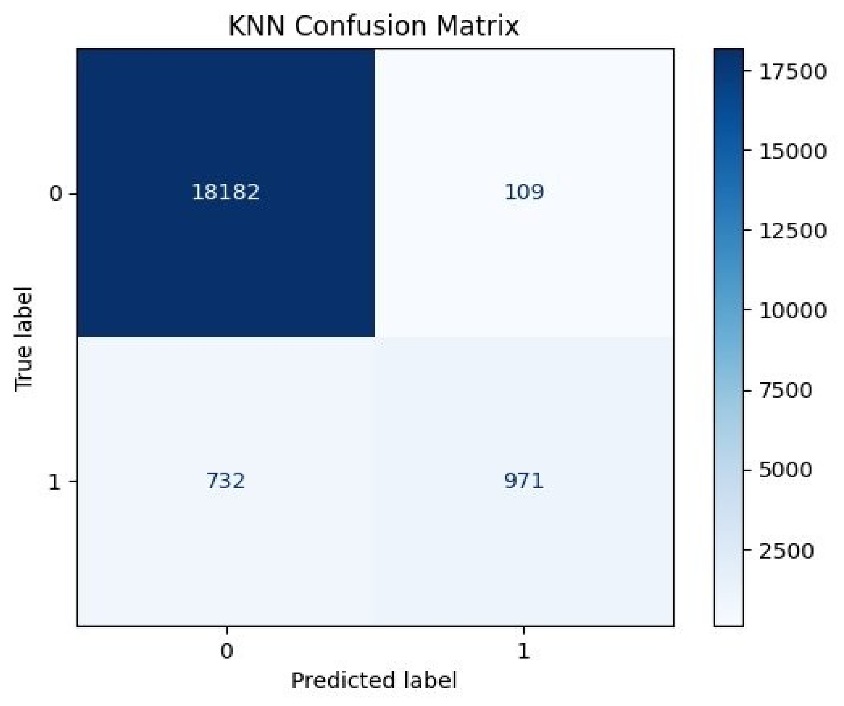
4. Different Model Training — Random Forest

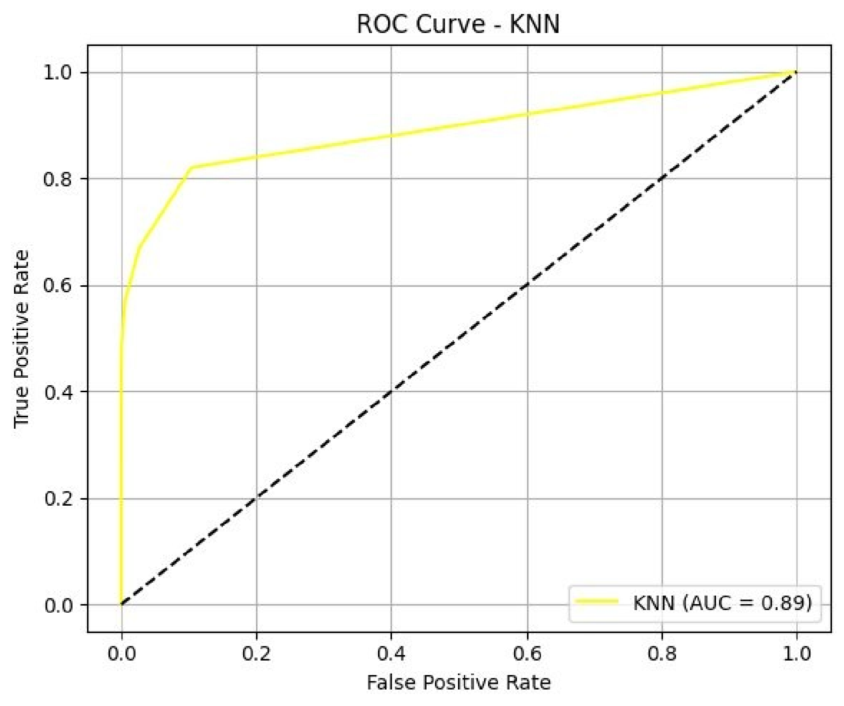




METHODOLOGY & RESULTS

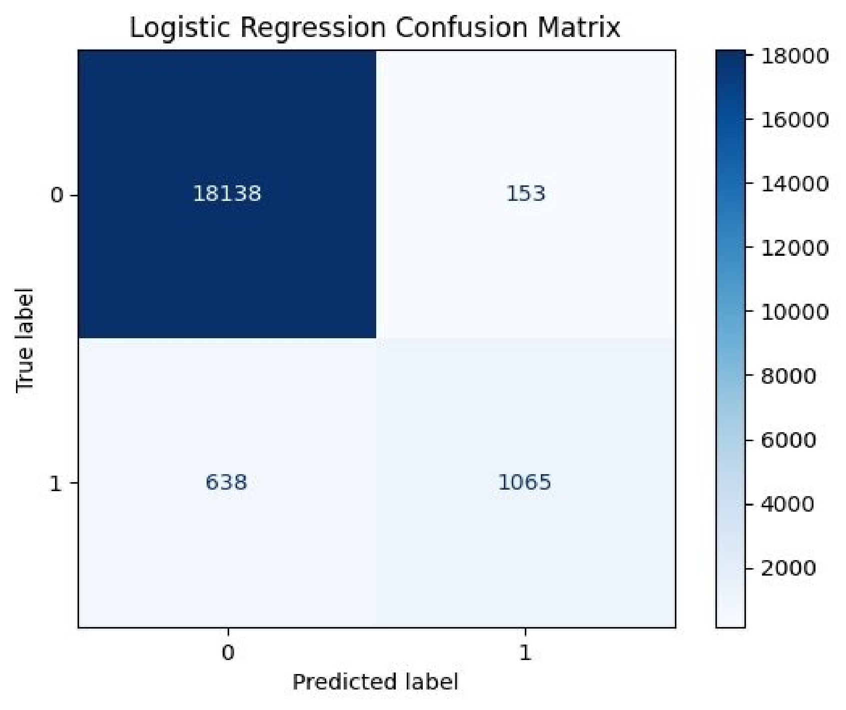
4. Different Model Training — KNN

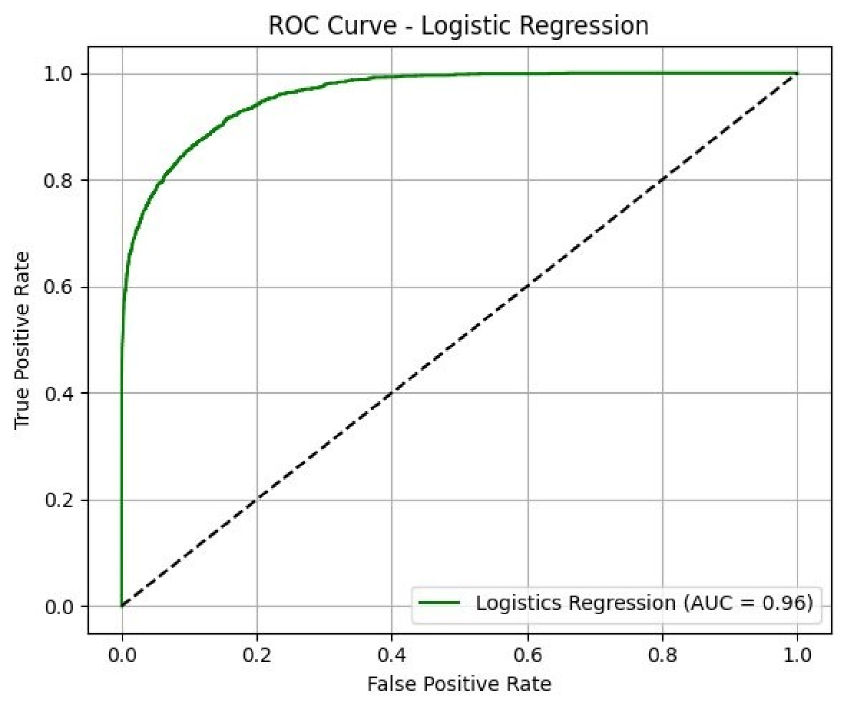




METHODOLOGY & RESULTS

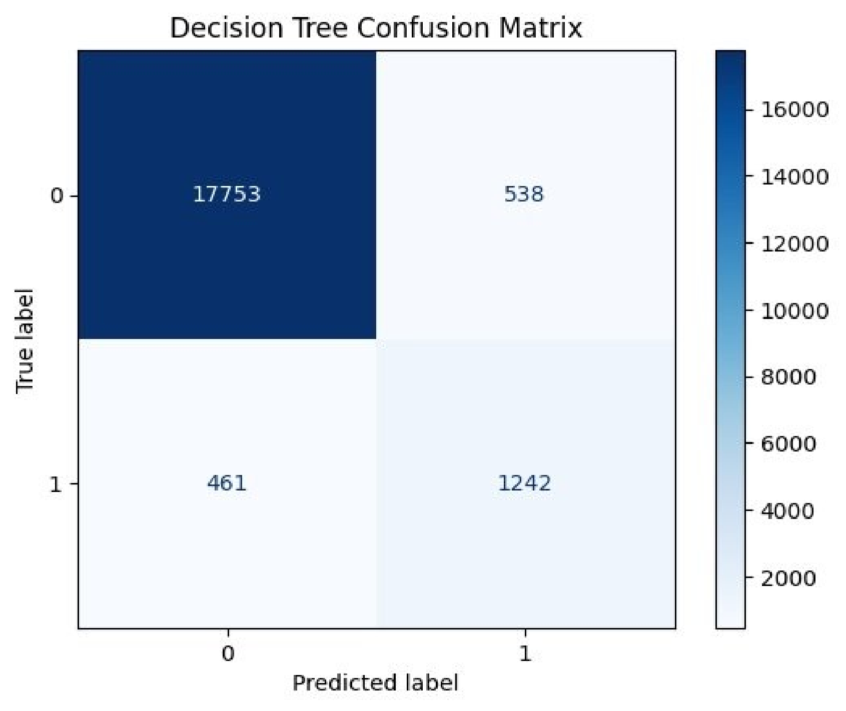
4. Different Model Training — Logistic Regression

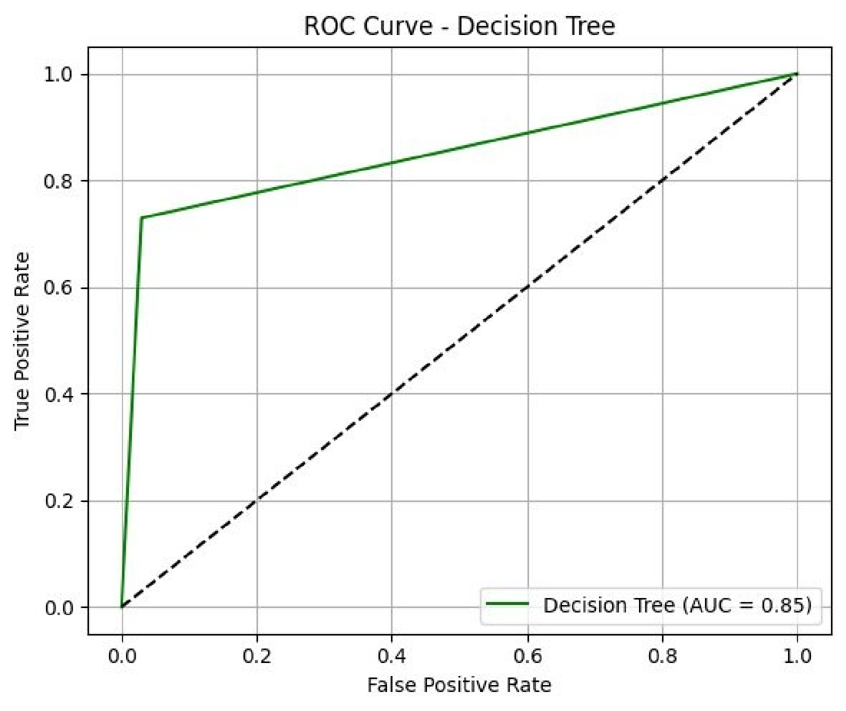




METHODOLOGY & RESULTS

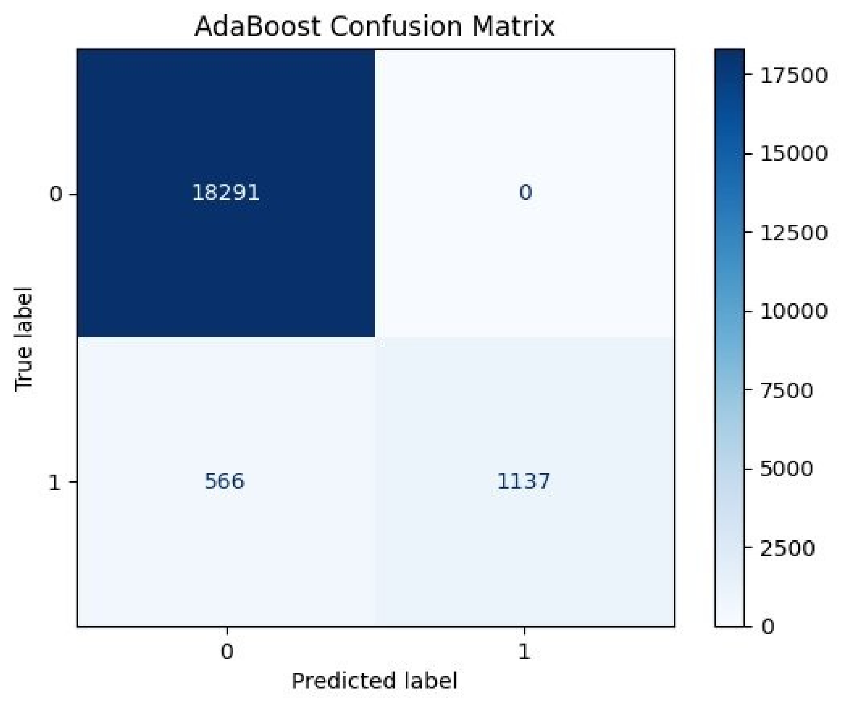
4. Different Model Training — Decision Tree

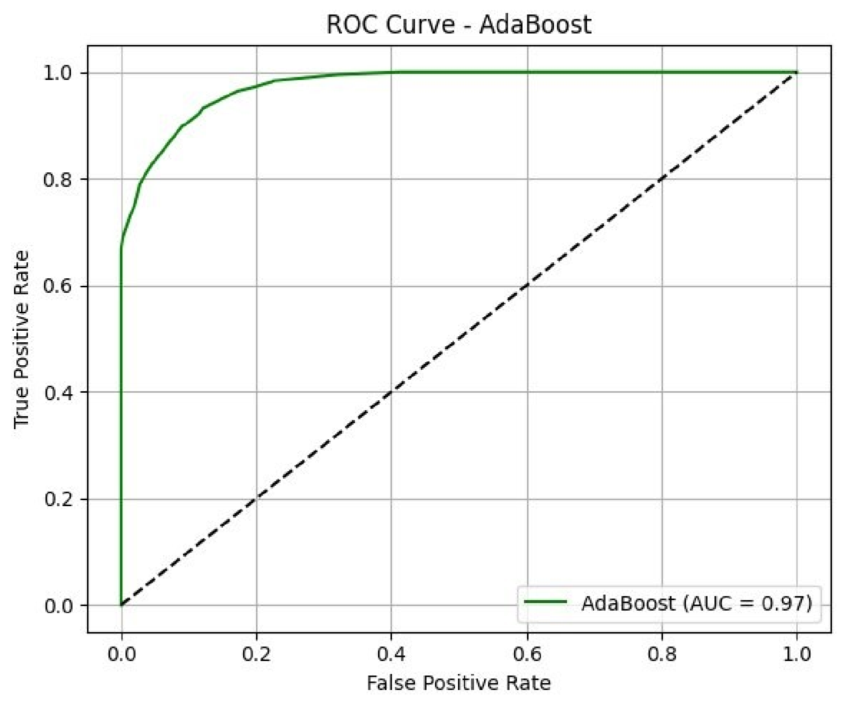




METHODOLOGY & RESULTS

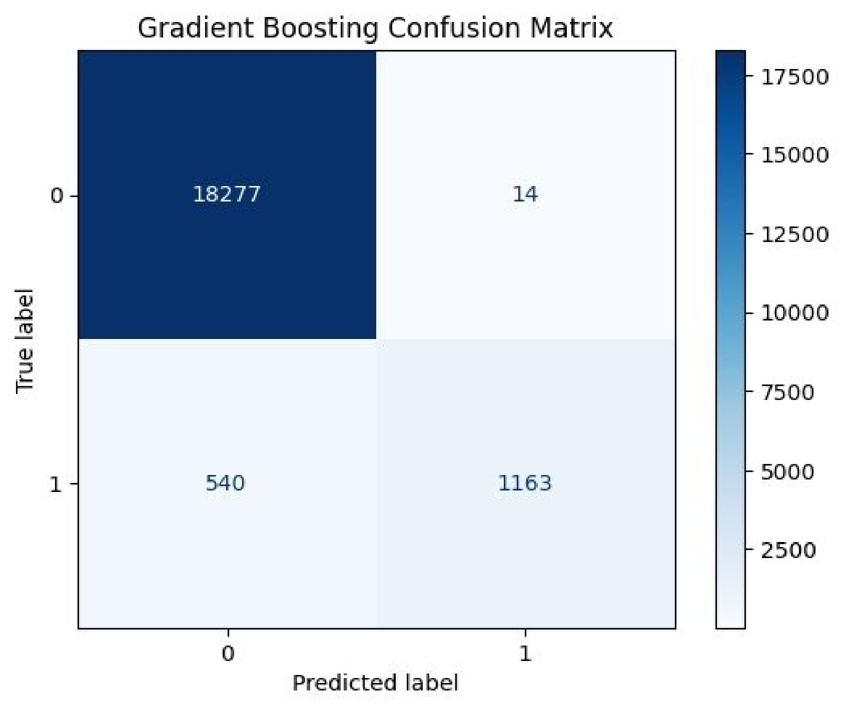
4. Different Model Training — AdaBoost

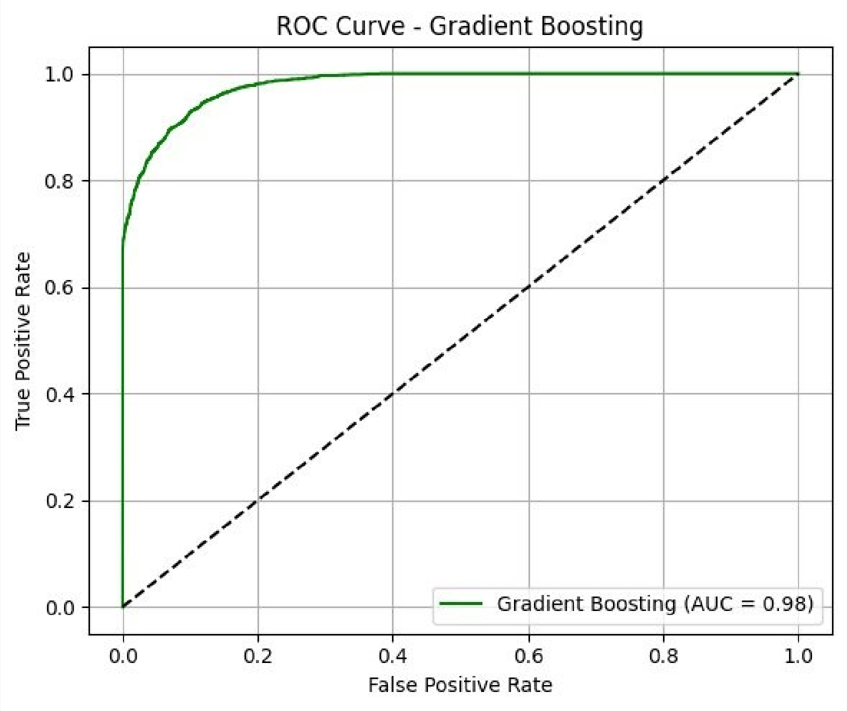




METHODOLOGY & RESULTS

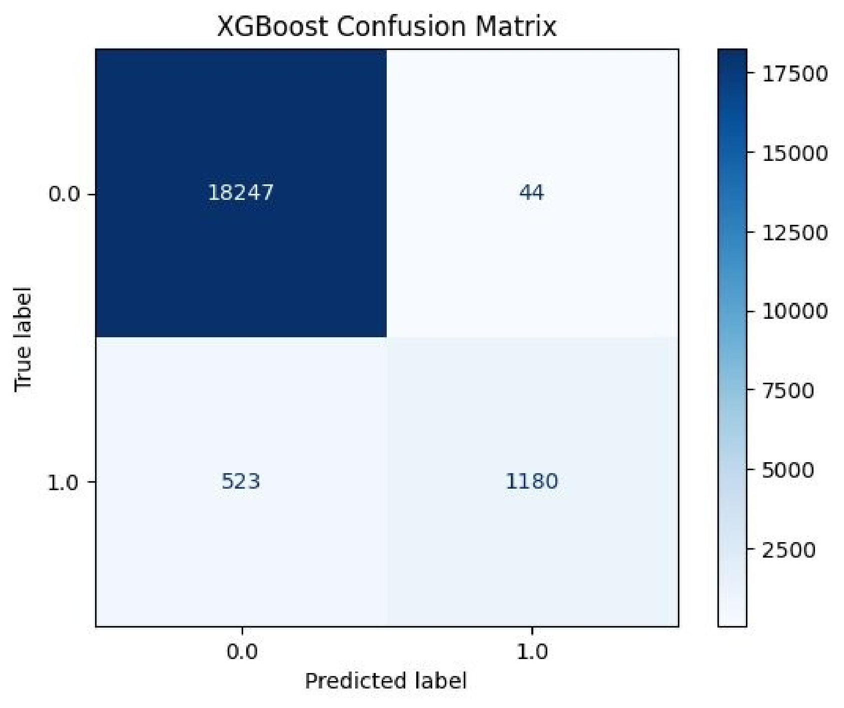
4. Different Model Training — Gradient Boosting

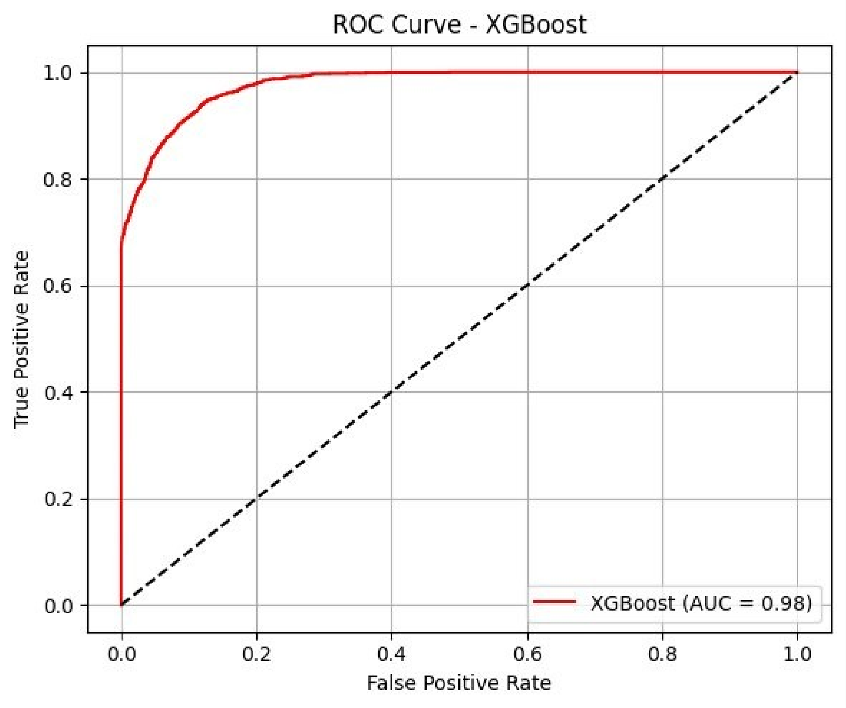




METHODOLOGY & RESULTS

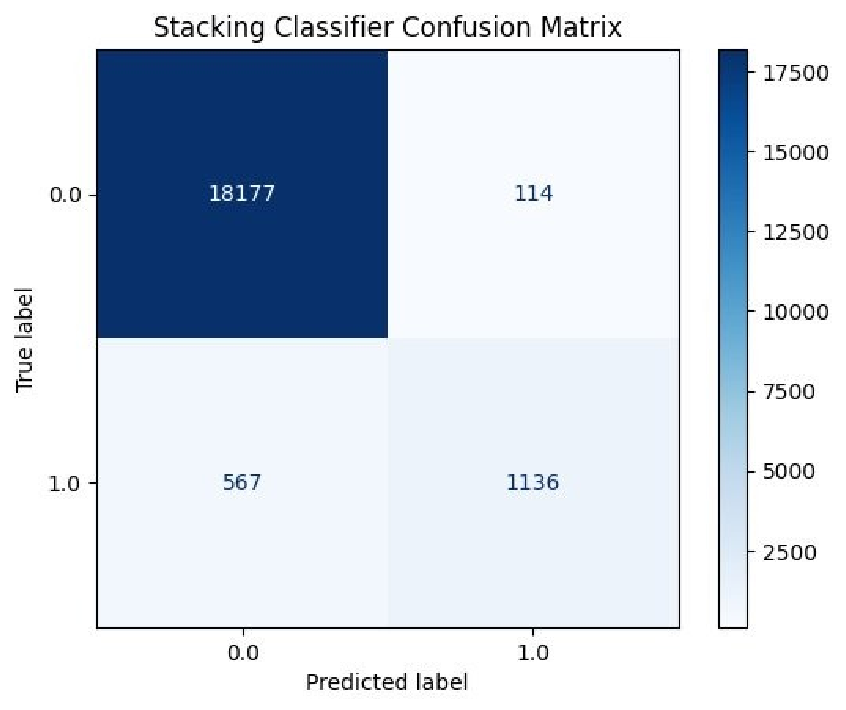
4. Different Model Training — XGBoost

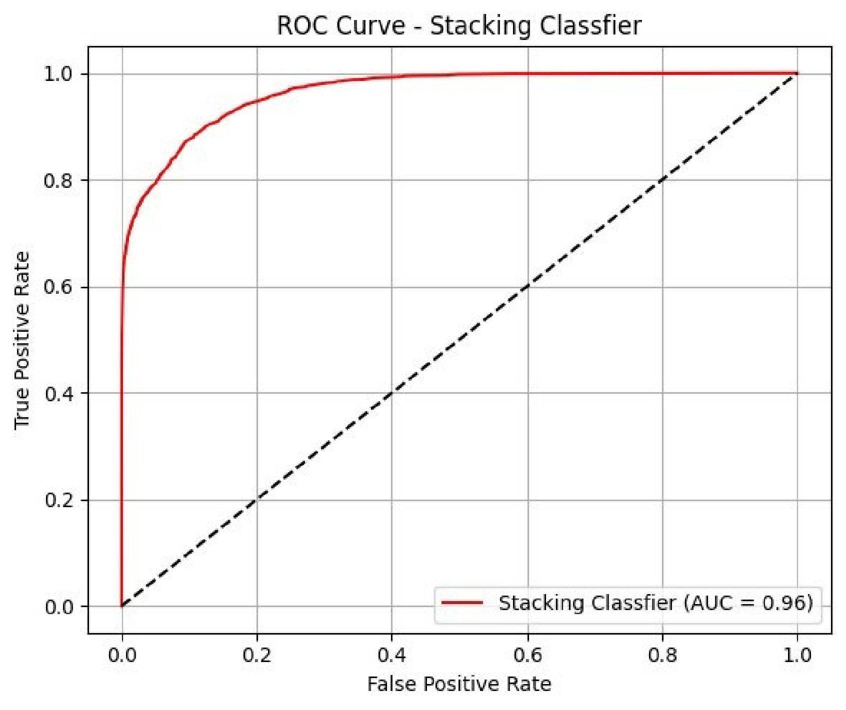




METHODOLOGY & RESULTS

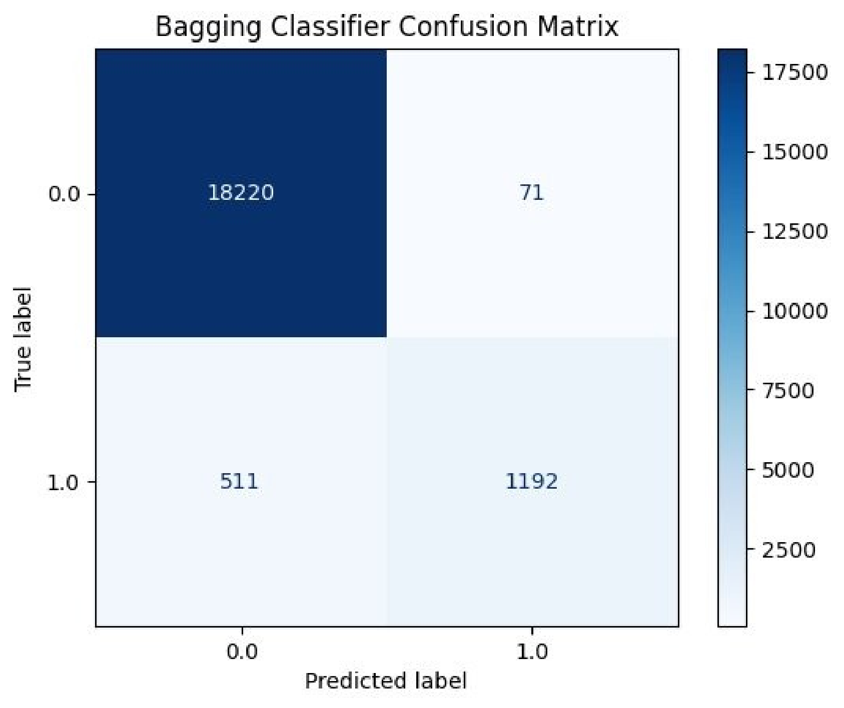
4. Different Model Training — Stacking Classifier

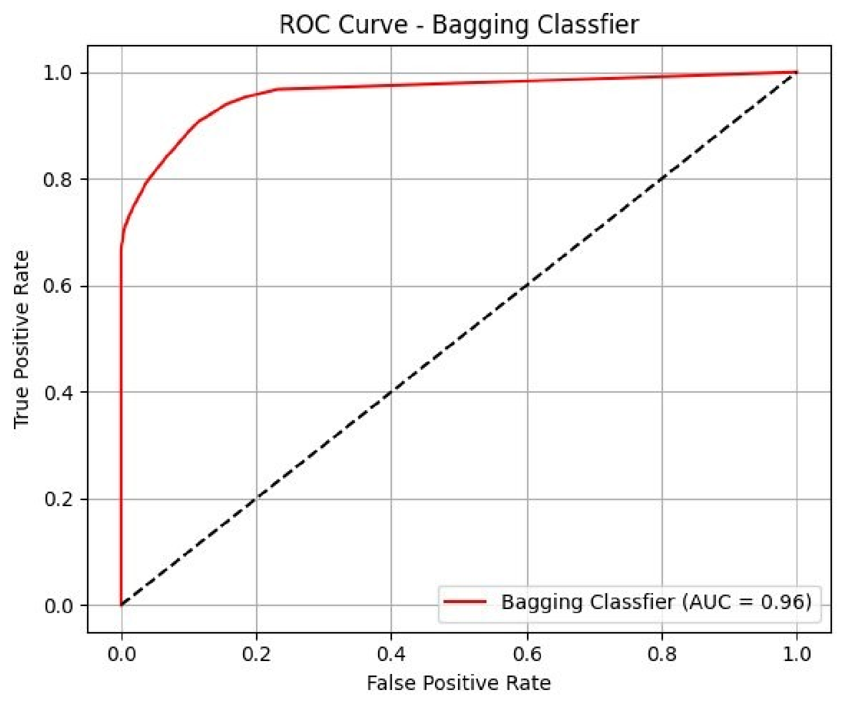




METHODOLOGY & RESULTS

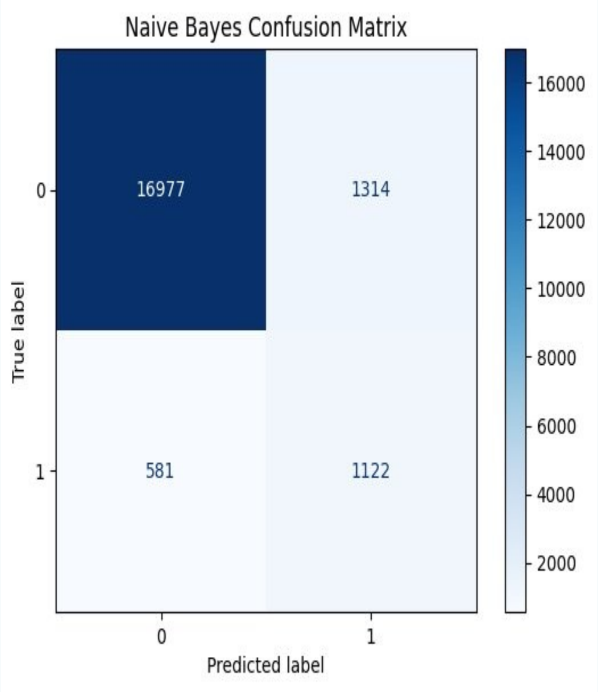
4. Different Model Training — Bagging Classifier



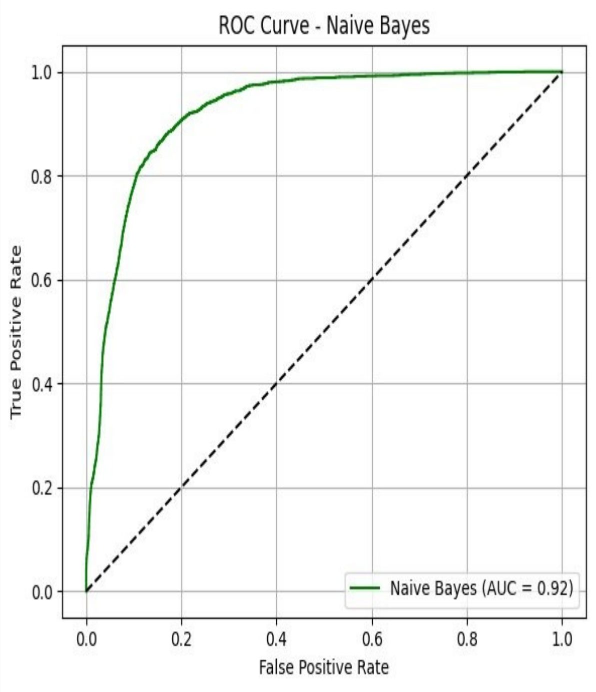


METHODOLOGY & RESULTS

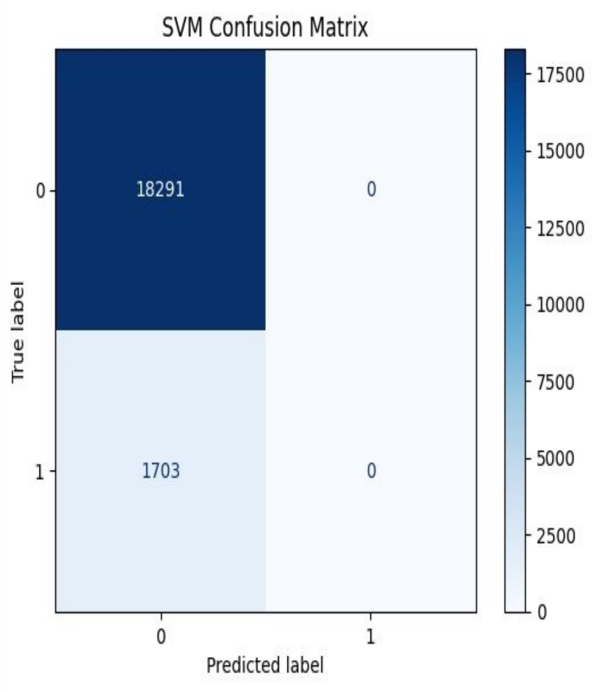
4. Different Model Training — Naive Bayes & SVM



Naive Bayes CM



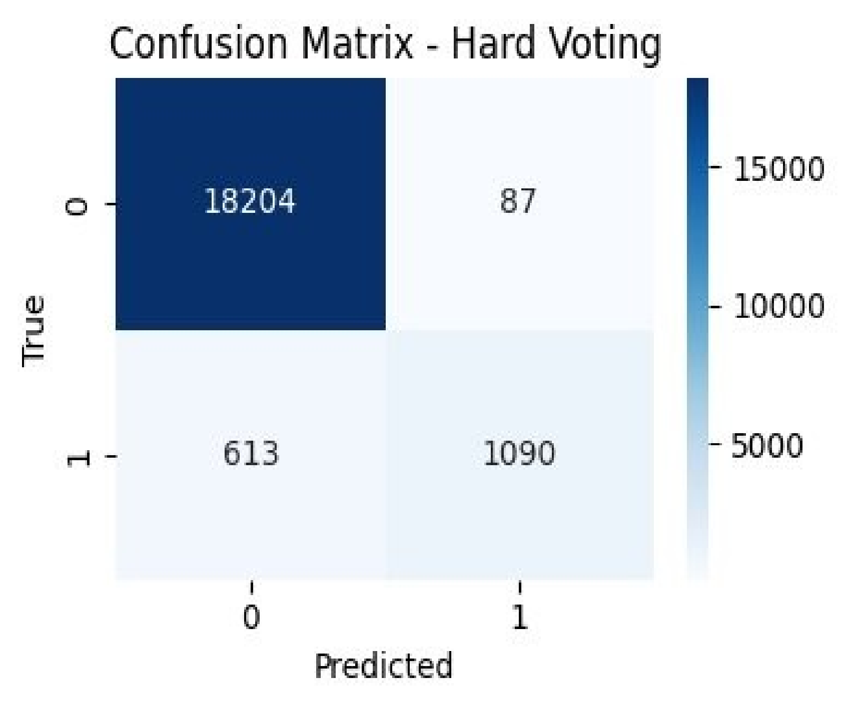
Naive Bayes ROC



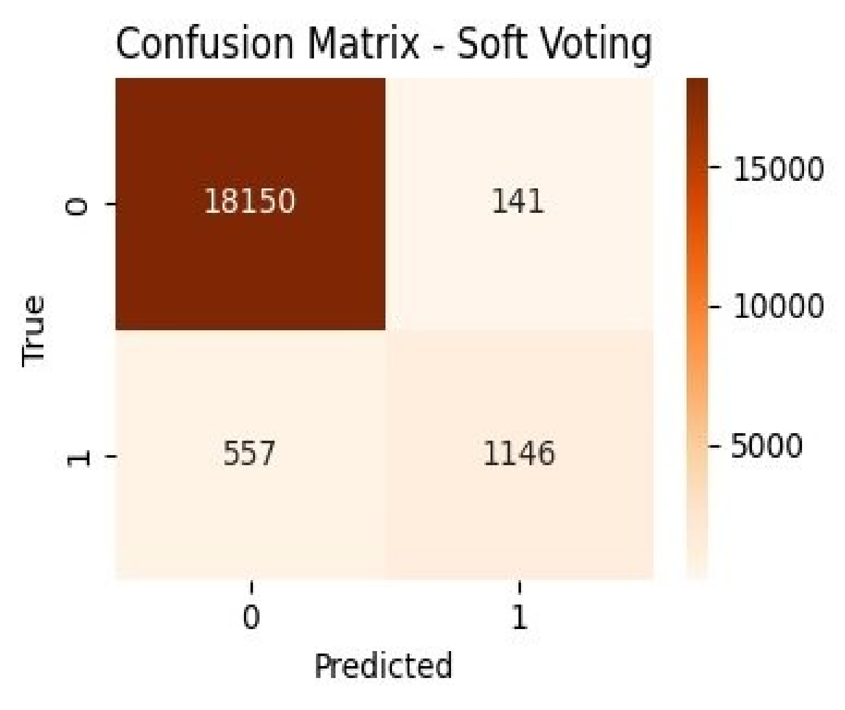
SVM CM

METHODOLOGY & RESULTS

4. Different Model Training — Hard & Soft Voting



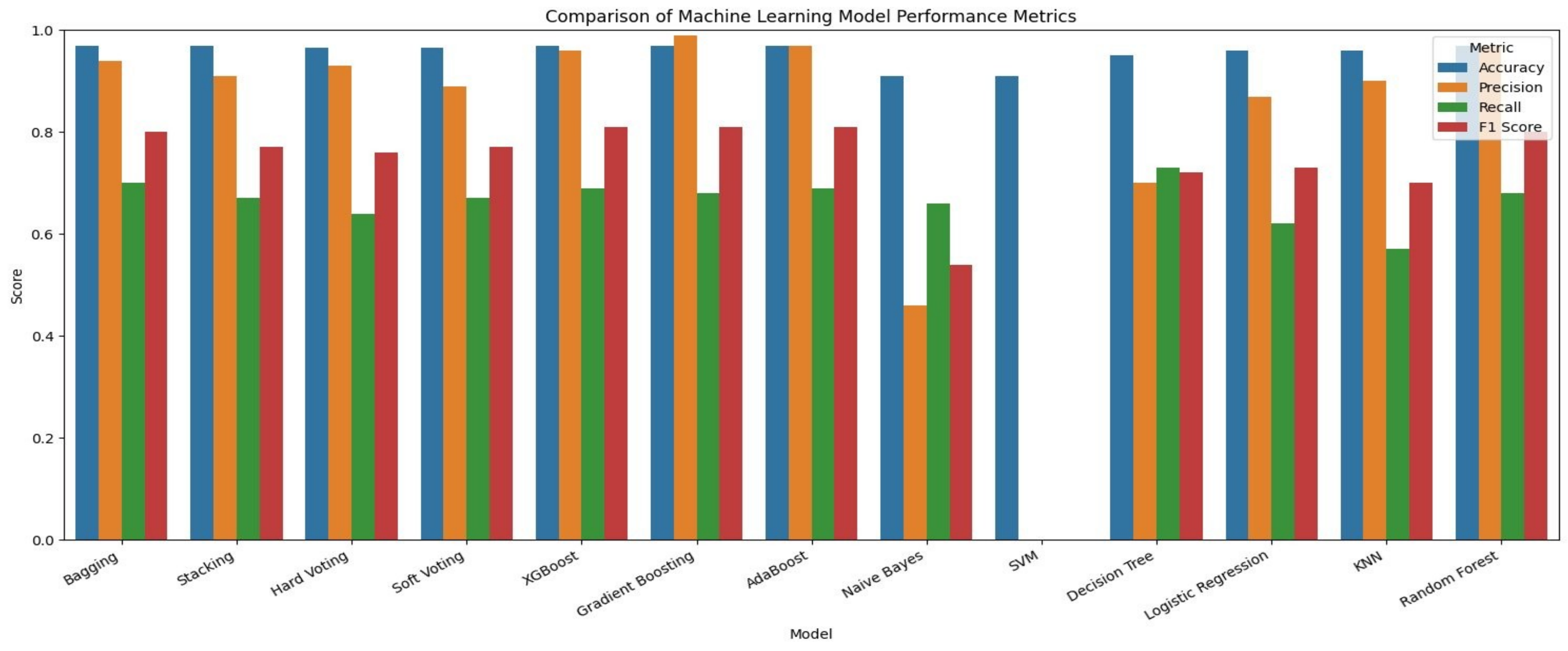
Hard Voting



Soft Voting

METHODOLOGY & RESULTS

5. Comparison of Machine Learning Model Performance Metrics



CONCLUSION

Best Models Overall

Gradient Boosting, AdaBoost & XGBoost all achieved F1 = 0.81 — best balance between precision and recall.

Precision vs Speed

Gradient Boosting/AdaBoost if false positives are costly. XGBoost for faster, scalable training.

Stable Alternatives

Bagging & Random Forest also performed well and offer stability and interpretability.

Key Predictor

HbA1c level & blood glucose were the most important features confirming their clinical significance.

Thank You

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